

From: Behrad Koohy behrad.koohy@luffy.ai

Subject: Project Kickoff Resources

Date: 13 February 2026 at 13:40

To: James Carlyle james.carlyle@soton.ac.uk, Ezra Daniels ed2g21@soton.ac.uk, Joshua Kilcoyne jk8g20@soton.ac.uk

Cc: Jamie Bayne jamie.bayne@luffy.ai

BK

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CAUTION: This e-mail originated outside the University of Southampton.

Hi all,

It was great to meet you all yesterday. As promised, we've put together some resources for this project.

The attached files include our plasticity rules and the project proposal. In addition to these, it might be worth having a look at the following literature:

- Chapters 1 & 2 of Hands-On Neuroevolution with Python by Iaroslav Omelianenko is a friendly introduction to neuroevolution.
- Evolving Neural Networks through Augmenting Topologies (<https://nn.cs.utexas.edu/downloads/papers/stanley.ec02.pdf>) is the paper that introduced NEAT.
- https://neuropsychics.ucsd.edu/courses/physics_171/Oja_1982.pdf introduces Oja's rules, a subset of the neuroplasticity rules that we are interested in.
- https://repository.lboro.ac.uk/articles/conference_contribution/Evolutionary_advantages_of_neuromodulated_plasticity_in_dynamic_reward-based_scenarios/9403988?file=17020676 is an example of neuroplasticity.

For implementations of NEAT, these look like promising avenues:

- <https://github.com/CodeReclaimers/neat-python/tree/master> seems like a well-maintained NEAT implementation.
- <https://github.com/google/brain-tokyo-workshop/tree/master/WANNRelease/prettyNEAT> is another option.

For environments, there are a number of control-based problems which could be quite interesting to explore. MuJoCo is usually the go to (or something like dm_control). This has the additional benefits of having lots of baselines for us to compare our results to. The Real-World RL Suite (https://github.com/google-research/realworldrl_suite) is a good example of an extension to MuJoCo/dm_control but it hasn't seen any updates for 6ish years. We're more than happy for you guys to explore other implementations of NEAT or environments as you see fit but would advise caution as you may run into some pretty nasty and frustrating versioning issues, particularly with older versions of Gymnasium/Gym/dm_control.

I'll send out an invitation for our next meeting in the next few days. If there are any questions or if we can provide any more clarity, please feel free to drop me an email or schedule a call!

Kind Regards,
Behrad

$$\begin{aligned}
\text{fixed}(s, \alpha, \beta) &= 0 \\
\text{Hebb}(s, \alpha, \beta) &= \alpha\beta \\
\text{Oja}_+(s, \alpha, \beta) &= \beta(\alpha - s\beta) \\
\text{Oja}_-(s, \alpha, \beta) &= \text{Oja}_+(s, \alpha, 1 - \beta) \\
\text{correlation}(s, \alpha, \beta) &= \alpha(2\beta - 1) \\
\text{anticorrelation}(s, \alpha, \beta) &= -\text{correlation}(\alpha, \beta)
\end{aligned}$$

where α is pre-synaptic activation, β is post-synaptic activation, and s is current synapse strength.

Update is computed as

$$\begin{aligned}
\Delta s^t &= \eta \text{rule}(s^t, \alpha, \beta) \\
s^{t+1} &= s^t + \Delta s^t
\end{aligned}$$

where η is the learning rate. η , initial strength s^0 , and choice of rule are evolved properties of each *synapse*.

You may find that clamping the synaptic strengths post-update helps stability.

Research_Project_PlasticityNE
AT_Proposal.pdf

